

Sec-A

Tutorial Sheet - 09

1. What is ISAM.

Ans ISAM stands for Indexed Sequential Access Method. It's a data access method used in database management system for organizing and accessing data records in a file. ISAM provides both sequential and indexed access to the data.

2. What are Blocks and Records?

Blocks are fixed-size units of storage on a disk, used for reading and writing data efficiently.

Records are collections of data fields within a database table, representing individual instances of data entities.

3. Types of File organization?

There are several types of file organization:

- (i) Sequential file organization - Data records are stored in sequential order based on a key field.
- (ii) Indexed Sequential Access Method (ISAM) - Combined sequential and indexed access methods.
- (iii) Hashed file organization - Data records are distributed across fixed-size buckets using a hashing function.
- (iv) B-Tree and B+ Tree - These are tree-based structures used for indexing data in databases.
- (v) Clustered file org. - Data records with similar values for one or more fields are physically stored together, typically based on a clustering index.
- (vi) Heap file org. - Records are stored in no particular order.

and the file may become fragmented over time.

4. What is data Mining?

Ans Data Mining is defined as extracting information from huge sets of data.

OR

Data Mining is the process of discovering meaningful patterns, correlations, and insights from large datasets.

5. What is data Warehouse?

Ans A data warehouse is a centralized repository that stores structured, historical data from multiple sources.

OR

A data warehouse is a subject oriented, integrated, time-variant and non-volatile collection of data. This data helps analysts to take informed decisions in an organization.

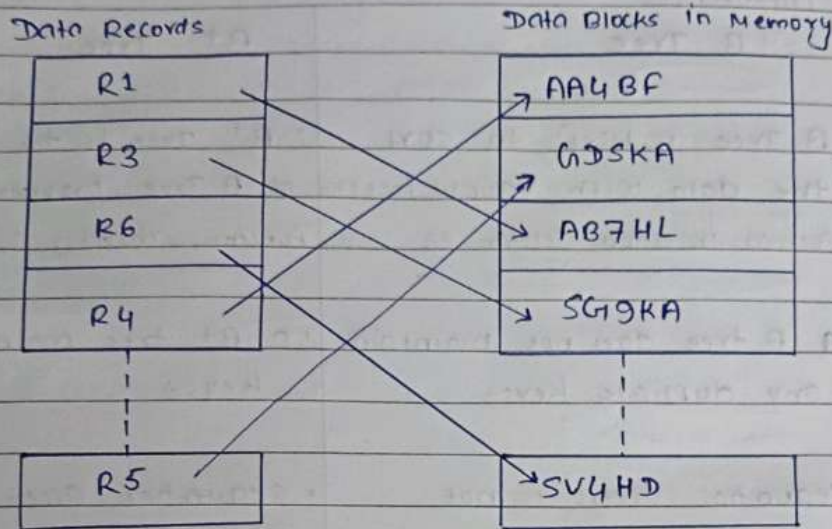
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1. What is Hash File Organization?

Ans Hash File organization is a method of organizing data in a database where records are distributed across a fixed number of buckets based on a hashing function. This function takes a key value associated with each record and calculates a hash code, which determines the bucket where the record will be stored.

Hash File organization uses the computation of hash function on some fields of the records. The hash function's output determines the location of disk block where the records are to be placed.

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When a record has to be received using the hash key columns, then the address is generated, and the whole is retrieved using that address. In the same way, when a new record has to be inserted, then the address is generated using the hash key and record is directly inserted. The same process is applied in the case of delete and update.

In this method, there is no effort for searching and sorting the entire file. In this method, each record will be sorted randomly in the memory.

Q2. Difference betⁿ B-Tree and B⁺ Tree.

Ans B-Tree: B-Tree is known as a self-balancing tree as its nodes are sorted in the inorder traversal. In B-tree, a node can have more than two children.

B⁺ Tree: B⁺ tree is an enhanced version of B tree, designed to facilitate efficient insertion, deletion, and sequential access. A B⁺ tree can hold a large number of children per node.

• difference :-
B Tree

B+ Tree

- | | |
|---|---|
| (i) B Tree is used to sort the data with various facilities such as insertion, deletion, etc. | • B+ Tree is the modified version of B Tree providing all the functions smoothly and quickly. |
| (ii) A B tree does not maintain any duplicate keys. | • A B+ tree maintains duplicate keys. |
| (iii) Sequential access is not possible. | • Sequential access is possible just like linked list. |
| (iv) All leaf internal and leaf nodes have data pointers. | • Only leaf nodes have data pointers. |
| (v) Searching in a B tree can be slow as all keys are not located at the leaf. | (v) Searching in a B+ tree is faster as all keys are stored at the leaf nodes. |
| (vi) The process of deletion is complex in a B tree. | (vi) The process of deletion is simpler in a B+ tree. |
| (vii) Leaf nodes are not linked to each other. | (vii) Leaf nodes are linked to each other. |
| (viii) For a particular no. of entries the height of the B tree is larger. | • The height of the B+ tree is lesser than B-tree for the same no. of entries. |
| (ix) B-Trees used in databases, search engines. | • B+ tree used in Multilevel indexing, database indexing. |

3. Objective of File organization.

- Ans
- It contains an optimal selection of records i.e., records can be selected as fast as possible.
 - To perform insert, delete or update transaction on the records should be quick and easy.
 - The duplicate records cannot be induced as a result of insert, update or delete.
 - For the minimal cost of storage, records should be stored efficiently.

There are many other objectives include:

- Efficient data Retrieval
- Effective data storage
- Data integrity
- Concurrency control
- Ease of maintenance
- Support Query Processing
- Indexing and searching
- Scalability and performance
- Data security

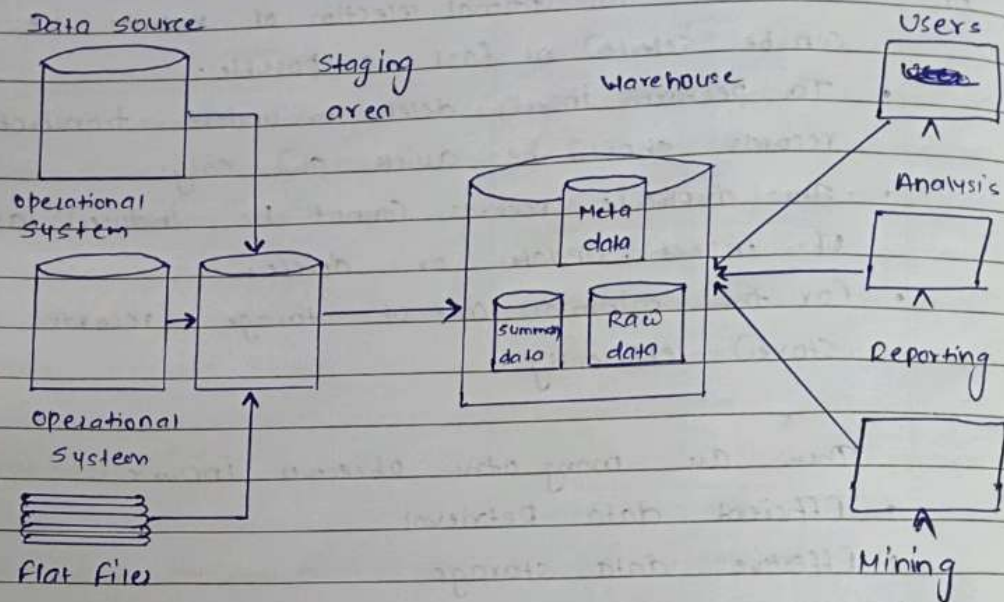
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1. Explain data warehouse Architecture.

Ans A data warehouse is a heterogeneous collection of different data sources organised under a unified schema.

Data Warehouse Architecture refers to the design and organization of the various components that make up a data warehouse system. It essentially defines how data flow throughout the system, from its origin in various sources to its final use in generating reports and analytics.

Architecture of data warehouse -



Components :

- **Source layer:** This layer encompasses the various Operational Systems and external data sources that provide the raw data for the data warehouse. eg: include CRM system, ERP systems, social media.

- **Staging Area:** This temporary storage area serves as a landing zone for the raw data extracted from source systems. Here, data goes through cleansing, validation, and transformation processes to prepare it for loading into the data warehouse.

- **Data Warehouse:** This is the central repository that stores the integrated and transformed data from various sources. This data warehouse is subject-oriented meaning it's organized around specific business subjects like sales, customers, or products.

- **Data Mart:** Data marts are departmental or functional subsets of the data warehouse, catering to the specific needs of a particular business area. They offer focused data for departmental analysis.
- **Analytical Tools:** These are software applications that allow users to query, analyze, visualize, and report on the data stored in the data warehouse or data marts.
- **OLAP (Online Analytical Processing) server:** OLAP servers provide tools and functionality for multidimensional analysis of data stored in the data warehouse.

Types of Architecture -

There are two main architecture approaches for data warehouses.

- **Three-tier architecture** - This is the most common design, consisting of separate tiers for the source data, the data warehouse, and the analytical layer. It offers a well-organized data flow and facilitates data management.
- **Single-tier Architecture** - This less common approach combines all the functionalities into a single layer. It's suitable for smaller data volumes but may not scale well for complex data warehouses.

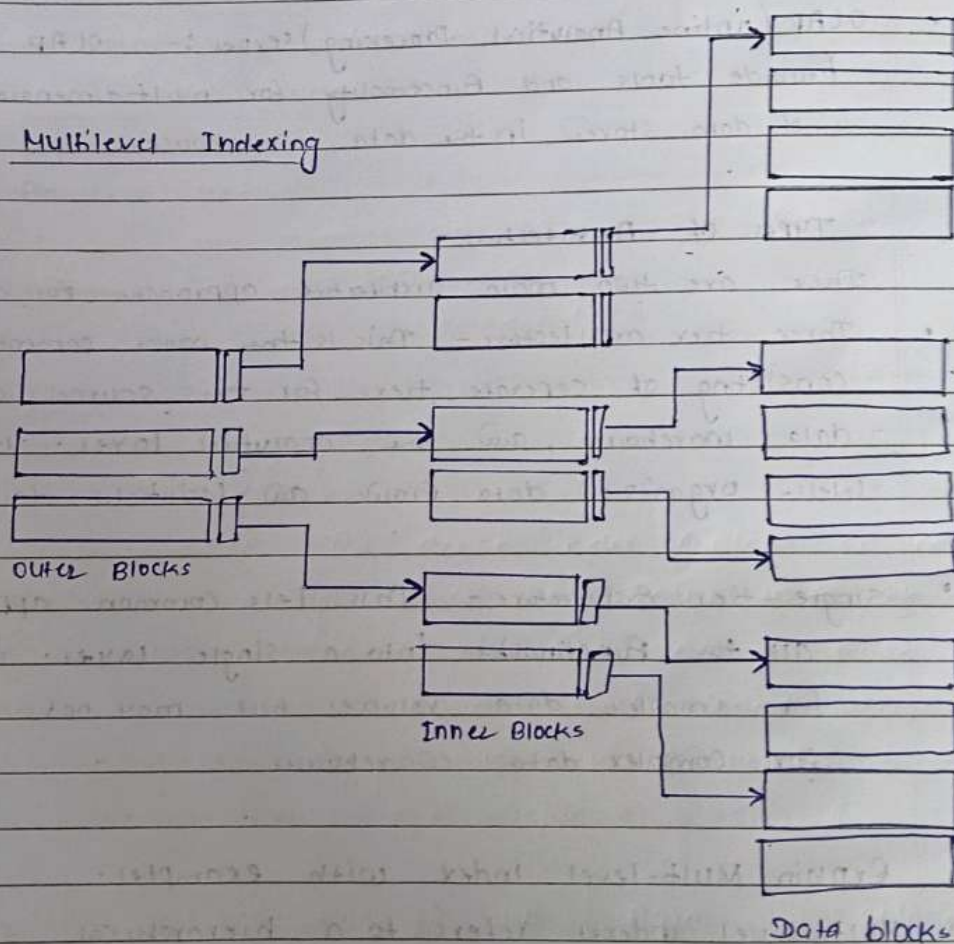
2. Explain Multi-level index with example.

Ans Multi-level index refers to a hierarchical structure of indexes. In this, each level of the index provides a more detailed reference to the data. It allows faster data retrieval, reduces disk access, and improves query performance. Multilevel indexes are essential in large

databases where traditional indexes are not efficient.

A Multi-level index can be created for any first-level index (primary, secondary or clustering) that has more than one disk block. It's like a search tree but adding or removing new index entries is challenging because every level of the index is on ordered file.

Multi-level Indexing



• Example: Sales Data Analysis

Let's say you have a large dataset containing sales information. It includes details like product category (electronics, clothing), product name (TV, shirt), region

(North, south), and Sales Figures.

So, A Multilevel index could have the following structure:

- Level 1: Product Category (electronics, clothing)
- Level 2: Region (North, south)
- Level 3: Product Name (TV, shirt)

With this structure we can quickly locate all electronics sales records for both North and South regions without sifting through the entire dataset.

Q 3. Explain indexing in detail. What is dense and sparse index.

Ans indexing is used to optimize the performance of a database by minimizing the number of disk access required when a query is processed. The index is a type of data structure. It is used to locate and access the data in a database table quickly.

Indexing is a technique used in database systems to improve the efficiency of data retrieval operations. It involves creating an additional data structure, called an index, which provides a mapping between the values of certain attributes (keys) and the physical location of the corresponding data records or disk blocks.

By using an index, the database system can quickly locate records that match specific search criteria without having to scan the entire dataset.

Index Structure :

Search key	Data
	Reference

- The First column of the database is the search key. that contains a copy of the primary key or candidate key of the table.
- The Second column of the data is the data reference. it contains a set of pointer holding the address of the disk block where the value of the particular key can be found.
- Indexing Methods:
 - Ordered indices → ordered indices is of two types.
 - Primary index → Dense index (i)
→ Sparse index (ii)
 - Clustering index
 - Secondary index

(i) Dense index: The dense index contains an index record for every search key value in the data file. It makes searching faster, but requires more space to store index records itself. Index records contain search key value and a pointer to the actual record on the disk.

UP	•	→	UP	Agra	1,604,300
USA	•	→	USA	Chicago	2,789,378
Nepal	•	→	Nepal	Katmandu	1,456,634
UK	•	→	UK	Cambridge	1,360,364

- (ii) Sparse index: In sparse index, index records are not created for every search key. An index record here contains a search key and an actual pointer to the data on the disk. To search a record, we first proceed by index record and reach at the actual location of the data. If the data we are looking for is not where we directly reach by following the index, then the system starts sequential search until the desired data is found.

china	•	→	China	Beijing	3,705,386
Russia	•	→	Canada	Ottawa	3,855,081
USA	•	→	Russia	Moscow	6,592,735
		→	USA	Washington	3,718,691

- Difference betⁿ. both :-

Dense index	Sparse index
(i) The index size is larger.	(i) index size is smaller.
(ii) Time to locate data in index table is less.	(ii) Time to locate data in index table is more.
(iii) Records in dense index need not to be clustered.	(iii) In case of sparse index, records need to be clustered.
(iv) indices are faster.	(iv) indices are slower.
(v) Data pointer in dense index point to each record in the data file.	(v) In sparse index, data pointer point to fewer records in data file.